HydroLoop - integrated simulation report

Generated 2026-10-14 | scenario: Reference | period: annual | climate: NOAA 1991-2020 normals + ASHRAE/FAO-56/Duffie-Beckman derivation

All results are SIMULATED and ESTIMATED; Subject to Field Validation. CO2 values are Gross Biological Fixation (not offsets); O2 values are stoichiometric (not a change in city air).

What limits HydroLoop

Bio loop	Carbon Limited (58% of lit hours)
Evaporative Cooling Limited	water quantity limits the media in 73% of active hours

Results with confidence

Condensate recovered	15,049.48 L | Low
Water used in reactor (from condensate)	1,540.36 L | Low
Overflow reused for condenser cooling (evaporated)	12,573.51 L | Low
Biomass produced	72.70 g | Low
Gross Biological CO2 Fixation	133.19 g CO2 | Low
Gross O2 produced (stoichiometric)	134.78 g O2 | Low
Condenser inlet air drop (mean, AC on)	1.45 K | Low
HVAC electricity saved (gross)	230.04 kWh | Low
PBR + control electricity	83.72 kWh | Low
Net electricity impact	47.80 kWh | Low
Net annual benefit	15.30 SAR/yr | Low
Net Carbon Impact	- | Not Yet Quantified

Validation targets

T1 Condensate covers 100% of daily reactor need + 20% margin	Pass: 100.0% of days meet it; overall coverage 100.0%
T2 Growth in conditioned condensate >= 80% of control	Warning: Not evaluable by simulation - requires the paired flask/panel experim
T3 Reactor temperature within 25-35 C for >= 90% of time	Fail: 45.3% of hours (max 48.0 C)
T4a No persistent decline in algae density (model)	Fail: longest run of declining days = 11 (after a 14-day start-up)
T4b No visible contamination	Warning: Not evaluable by simulation - requires observation/microscopy

Economics

Net annual benefit	15.3 SAR/yr
Allowable CAPEX for payback within evaluation period	76 SAR (5 yr)
Simple payback	not computable (CAPEX incomplete)
Biomass revenue	0 (no documented product/market/price)

Key assumptions (Design Inputs to replace with measurements)

External shading fraction on panel	0.0 -
Evaporator air flow per TR	400.0 cfm/TR
Outdoor-air fraction at coil inlet	0.1 -
Indoor (return) relative humidity	55.0 %
Coil leaving air dry-bulb	13.0 C
Coil leaving air relative humidity	95.0 %
Load model: balance temperature (RTF = min_rtf)	18.0 C
Load model: temperature at RTF = 1	44.0 C
Load model: minimum runtime fraction	0.3 -
AC COP at 35 C outdoor (rating point)	3.2 W/W
Evaporating temperature (Carnot-fraction cross-check)	7.0 C
Condensing temperature minus condenser inlet air	12.0 K
Condenser air flow per TR	1000.0 cfm/TR
Condenser fan total efficiency	0.3 -
Minimum orifice diameter before clogging risk	0.002 m
Panel wall light transmittance	0.9 -
Panel solar absorptance (thermal)	0.8 -

Heat-loss coefficient per face	15.0 W/m2K
Small pump wire-to-water efficiency	0.25 -
Minimum pump electrical power	5.0 W
LED photon efficacy	2.5 umol/J
Fraction of photons above the validated PPFD credited to gro	0.5 -
Biomass-specific light absorption coefficient	0.15 m2/g
Thermotolerant profile Tmin	10.0 C
Ash fraction of dry weight	0.0 -
Bicarbonate carbon availability	1.0 -
Sensors power	1.5 W
Controller (ESP32) power	1.0 W
PV performance ratio	0.75 -

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[FLUIDS] Standard fluid-mechanics references (e.g., White F.M., Fluid Mechanics): sharp-edged orifice discharge coefficient ~0.61; Darcy-Weisbach with Swamee-Jain (1976) friction factor; typical minor-loss coefficients (entrance 0.5, standard 90 deg elbow ~0.9); smooth plastic pipe roughness ~0.0015 mm.
[PROJECT] HydroLoop / EcoFlow Algae Facade - Miyahthon 3 digital-bootcamp submission (validation targets).
[NONE] No reliable source found - value is a user Design Input and must be measured or replaced.

Warnings

- Site elevation: Jeddah coastal elevation; replace with surveyed value
- Monthly mean daily max temperature (Jan..Dec): Monthly means: real hourly extremes (e.g., 45-52 C records) are not represented; use NASA POWER/measured hourly data for extremes
- Monthly mean relative humidity (Jan..Dec): Mean RH converted to a constant daily humidity ratio at the daily-mean temperature (ASHRAE design-day convention); hourly humidity peaks are not represented
- Angstrom coefficient a_s: FAO-56 default for uncalibrated sites; Jeddah-calibrated coefficients or NASA POWER data should replace it
- Ground reflectance: Urban Jeddah surfaces may differ
- PAR fraction of global solar radiation: Typical 0.45-0.50; not measured for Jeddah
- Measured condensate rate while AC operates: Enter the measured value from the 3-day collection test
- AC nominal cooling capacity: Set to the actual target unit; Jeddah 2024 study did not report unit capacity
- Evaporator air flow per TR: Common HVAC design convention (~400 cfm/ton); use nameplate/measured flow
- Outdoor-air fraction at coil inlet: Split units are often recirculating with infiltration; measure or use Measured Mode
- Indoor (return) relative humidity: Indoor humidity in Jeddah buildings is load-dependent; measure it
- Load model: balance temperature (RTF = min_rtf): Simple linear bin-type load assumption, not a building load simulation
- AC COP at 35 C outdoor (rating point): Replace with the unit's nameplate/datasheet value (Manufacturer Data)
- Condenser air flow per TR: Use the outdoor-unit datasheet
- Compressor power reduction per K of condenser inlet air pre-cooling: Linearisation of one experiment; not a general law. Capacity gain (COP +21.4%) is NOT credited.
- Media saturation effectiveness: Effectiveness depends on face velocity and pad depth
- Media air-side pressure drop: Penalty applies whenever the pad is installed, wet or dry
- Air face velocity through the media: Lower velocity = larger pad area, lower pressure drop
- Pre-filter loss fraction: Set >0 only if measured
- Panel wall light transmittance: Use acrylic/polycarbonate datasheet value; fouling reduces it
- Panel solar absorptance (thermal): Must be calibrated with the 5-minute temperature log
- Heat-loss coefficient per face: Must be calibrated with measured panel temperature
- Culture pH (controlled/measured): pH is an input, not simulated (no carbonate chemistry model)
- Small pump wire-to-water efficiency: Use pump curve
- LED photon efficacy: Use LED datasheet
- Biomass yield on absorbed photons: Measured at <=930 umol/m2/s; outdoor peaks are higher
- Fraction of photons above the validated PPFD credited to growth: 0 = conservative, 1 = optimistic
- Biomass-specific light absorption coefficient: No verified value found during audit; measure absorbance vs dry weight
- Temperature response points UTEX 1230 (T: relative final dry weight): Ratios of final dry weight, not specific growth rates; below 25 C held

at 0.847 (no data at this light)
- Thermotolerant profile Tmin: No verified Tmin found; affects only cool months
- Thermotolerant profile Topt: Heterotrophic optimum used for a phototrophic system
- Suitable reactor temperature range - low (Target 3): Derived: relative growth >= ~0.7 for UTEX 1230
- Biomass carbon fraction: Literature assumption; replace with CHN measurement (Measured Input)
- Species stoichiometry factor (CO2 per g biomass): Not independent of the carbon-content method (same formula basis)
- Team reference: CO2 per g biomass (DOI 10.1016/j.biortech.2014.05.118): Could not be read from the original paper; 1.83 equals 0.50*44/12 exactly (Chisti 2007), so it is not an independent check
- Biomass elemental formula per C-mol (CHxOyNzPw): Generic microalgal formula, not measured for KAUST 111
- Ash fraction of dry weight: Measure ash; default 0 overstates C, N, P, O2 per g DW slightly
- Photosynthetic quotient (PQ method): High-precision range 1.0-1.36; nitrate assimilation pushes PQ to the upper end
- Ambient CO2 concentration: Background value; urban Jeddah air is likely higher; local measurement recommended
- Gas-to-liquid CO2 transfer (utilisation) efficiency: Measured for enriched gas; for ambient air it is a Design Input to measure (inlet/outlet CO2)
- DO level flagged as high: Not strain-specific; indicative only
- Air pump specific power: Rated at free flow; real back-pressure lowers flow
- Electricity tariff (government sector): ECRA 2018 decision via Saudi Gazette; confirm on the latest SERA/SEC tariff page before use
- CAPEX - Tank: Enter a dated quotation; no prices were invented
- CAPEX - Transparent reactor panel: Enter a dated quotation; no prices were invented
- CAPEX - Pipes: Enter a dated quotation; no prices were invented
- CAPEX - Valves: Enter a dated quotation; no prices were invented
- CAPEX - Filters: Enter a dated quotation; no prices were invented
- CAPEX - Circulation pump: Enter a dated quotation; no prices were invented
- CAPEX - Air pump: Enter a dated quotation; no prices were invented
- CAPEX - Sensors: Enter a dated quotation; no prices were invented
- CAPEX - Controller: Enter a dated quotation; no prices were invented
- CAPEX - Evaporative media: Enter a dated quotation; no prices were invented
- CAPEX - Drain pan: Enter a dated quotation; no prices were invented
- CAPEX - Solar PV: Enter a dated quotation; no prices were invented
- CAPEX - Installation: Enter a dated quotation; no prices were invented
- CAPEX - Structure/frame: Enter a dated quotation; no prices were invented
- Biomass price (documented product & market only): Revenue stays 0 unless product, market and a dated price source are entered
- Jeddah split AC annual condensate: Do not scale to a specific tonnage